

DYNAMICS REPORT

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ABSTRACT

This paper presents a comparative technical analysis of 4 vibration measurement technologies:

- Piezoelectric accelerometers
- MEMS accelerometers
- Laser Doppler Vibrometers (LDVs)
- Strain gauge accelerometers

For each, key parameters such as frequency bandwidth, sensitivity, dynamic range, mass, cost and environmental sustainability are evaluated to determine their advantages and limitations. The relationship between the sensing mechanism and measurement performance is examined for each technology, particularly with respect to high-frequency dynamic response, static acceleration measurement, and non-contact operation. A practical application- an automotive engine- is used to justify sensor selection. The findings proved piezoelectric accelerometers to have the optimal set of parameters for the task, providing the best balance of bandwidth, dynamic range, robustness and practicality for the application.

INTRODUCTION & LITERATURE REVIEW

Vibration measurement is a critical aspect of modern engineering, with applications spanning automotive performance analysis, machine condition monitoring, and structural analysis. Accurate detection and quantification of vibratory motion allows engineers to identify mechanical faults, optimise performance, and prevent failures. As industrial and research applications increasingly demand high-bandwidth, and sometimes non-contact measurements, a variety of sensing technologies have been developed, each with unique advantages and limitations. The following sections provide an in-depth analysis of the theories, properties and mechanisms of the 4 technologies named above, evaluating parameters and discussing their suitability for practical applications.

Piezoelectric accelerometers

A piezoelectric accelerometer is a common sensor used for measuring acceleration, vibration and dynamic motion by using the piezoelectric effect, also called piezoelectricity.

Piezoelectricity is the ability of a material to generate an electrical charge in response to a mechanical stress. This arises from the relationship between crystal symmetry and electric dipole moments, only occurring in materials whose crystal structures lack a centre of inversion symmetry. Inversion symmetry means for every atom in the crystal structure, there is an identical atom with coordinates $(-x, -y, -z)$. This symmetry requirement is essential- out of the 32 crystal classes, 21 are non-centrosymmetric, and 20 of these display piezoelectricity (except the cubic class 432) [1]. In

non-centrosymmetric crystals, mechanical stress displaces ions, shifting the centres of the positive/negative charges and produces a net electric dipole moment. Collectively, all of these dipoles generate a polarisation, which mathematically is proportionate to stress:

$$D_i = d_{ij}T_j \quad (1)$$

Where D_i is the electric displacement, T_j is mechanical stress and d_{ij} represents the piezoelectric coefficient tensor.

When the accelerometer undergoes acceleration, an inertial force is generated according to Newton's Second Law ($F = ma$), where m is the seismic mass. The force is then transmitted to the piezoelectric element as a compressive or shear stress. Using the electric displacement equation above (1), when integrating over the electrode area we get a charge output proportional to applied force:

$$Q = dF$$

Substituting in $F = ma$ gives:

$$Q = dma \quad (2)$$

Therefore proving that charge is proportional to acceleration. The charge output is usually converted into a measurable voltage using a charge amplifier- the resulting voltage signal is therefore also proportional to the input acceleration [2].

Due to the high mechanical stiffness of piezoelectric materials, these accelerometers have high dynamic range and wide frequency bandwidth. Dynamic range is the range of acceleration magnitudes it can measure linearly- having a high dynamic range of $>120\text{dB}$ makes them well suited to vibration measurement [3].

Frequency bandwidth is the range of frequencies over which they can accurately measure or respond to a signal- piezoelectric accelerometers typically operate between frequencies of $1\text{Hz}-20\text{kHz}$. They can't measure true DC (0Hz) acceleration due to the slow internal leakage of generated charge. This means they are ideal for measuring dynamic acceleration but aren't suitable for static or very low frequency measurements e.g. gravity. This means they are most commonly used in applications such as machine condition monitoring, structural analysis and impact testing- all uses that require high frequency measurements.

Sensitivity is how much electrical output that is measured per acceleration, usually around $\approx 100\text{mV/g}$, depending on which specific type of accelerometer you use. The high sensitivity gives it a better ability to detect small accelerations, but this is also dependent on dynamic range and frequency bandwidth. Similarly, temperature operating limits also vary, with common industrial accelerometers having a range of between -50°C and 120°C , and special sensors having up to $400^\circ\text{C}+$ in some cases. Sensitivity and bandwidth can shift slightly with temperature, due to the piezoelectric crystals properties and the stiffness of the material changing. This can cause slightly more or less voltage being outputted for the same

acceleration.

Despite this, they are very robust- their stiff internal construction means they can survive very high shock levels and don't wear out, making them perfect for use on systems like turbines, gearboxes and impact or shock measurement.

Piezoelectric accelerometers also come in a range of sizes, ranging from miniature types that weigh between 0.2-1g, up to high sensitivity types that can weigh up to 50-100g. Changing the mass affects the natural frequency of the accelerometer, as for a vibrating system:

$$f_n = \frac{1}{2\pi} \sqrt{\frac{k}{m}} \quad (3)$$

This means larger sensors are usually more mechanically robust with a higher sensitivity, but measure a lower frequency bandwidth. Smaller sensors are more delicate with a lower sensitivity, but have a higher bandwidth, making them ideal in lab/test environments. The size of the sensors also affects the price, with larger sensors being more expensive than smaller ones.

However, for situations needing tilt sensing or orientation, MEMS accelerometers are used instead.

MEMS accelerometers

An MEMS accelerometer is a type of sensor that measures acceleration by converting mechanical motion into an electric signal.

Micro-electro-mechanical systems (MEMS) are micro-devices that integrate mechanical components and electronic circuits on a single chip. Despite their size, they can contain mechanically moveable elements such as springs, beams or suspended masses [4] This integration of systems within a single device enables compact, low power and cost-effective sensing.

MEMS accelerometers usually consist of a proof mass suspended by structures such as springs within the MEMS chip. The proof mass is a small internal mass inside the accelerometer that moves when the device experiences acceleration. This is because under acceleration, the inertial force displaces the proof mass- as the stiffness of the suspension and the mass are known, the displacement can be related quantitatively to acceleration.

Because of the suspended mass, they are able to measure tilt and orientation [5]. If acceleration is constant:

$$x = \frac{ma}{k} \quad (4)$$

Where k = stiffness and m = mass. This means the displacement is measurable, making it perfect for measuring tilt and orientation in devices such as phones and machinery. It also makes it able to measure DC (0Hz), but has a bandwidth limited to low kHz due to the proof mass being tiny. Low power-consuming accelerometers usually have a range of 0 to 50-200Hz, with high-performance MEMS having a range of 0 to 2-5kHz.

Sensitivity of MEMS varies depending on measurement range. Low range accelerometers (~650mV/g) are very sensitive, making them ideal for tilt and gesture detection. High range accelerometers (<~82mV/g) have low sensitivity and are used in motion capture and automotive crash detection.

Their dynamic range (~60-100dB) also enables them to measure very small human motion and gravitational tilt, which combined with their very small size and mass (usually around 2-5mm² weighing milligrams) makes them perfect for wearables and smartphones. Their operating temperature range, which being between -40°C-85°C, with a low mass and cost, also makes them ideal for this use.

Typically, they're manufactured on silicon substrates using batch processing. Micromachining techniques (such as surface and bulk etching) produce the suspended structures and integrate them with electronics. This reduces size, power consumption and cost, again making it an ideal sensor for mass produced products such as vehicle systems and mobile phones.

Laser Doppler Vibrometers

A Laser Doppler Vibrometer (LDV) is a non-contact instrument used to measure surface velocity and displacement by applying the Doppler effect to laser radiation.

The Doppler effect describes the change in the observed frequency/wavelength of a wave due to relative motion between the source of the wave and the observer. When a source and observer move relative to one another along the line of sight, the spacing between successive wavefronts arriving at the observer is altered. If the source approaches the observer, wavefronts are compressed, increasing observed frequency and decreasing wavelength. If the wave source moves away, the opposite happens- wavefronts are stretched, wavelength increases and frequency decreases. This effect doesn't change the natural properties of the source, it simply alters how often wavefronts arrive at the observer.

An LDV operates by firing a monochromatic laser onto a vibrating surface. When the surface moves with velocity v , the frequency of the reflected light undergoes a Doppler shift proportional to the velocity component in the same direction as the laser beam. For reflected light:

$$f_D = \frac{2v}{\lambda} \quad (5)$$

From this, LDVs only measure velocity, not acceleration- this can be integrated to displacement or differentiated to acceleration, however the technology only measures velocity. Sensitivity is adjustable, ranging from 1 mm/s per Volt up to 100mm/s per Volt, depending on the selected scale.

They have a frequency bandwidth of DC (0Hz) to 20kHz/100kHz for basic/industrial systems, and because they have no mass/spring system they have a wide bandwidth, limited only by photodetector bandwidth and the processing speed of the electronics. [6] This makes them perfect for MEMS device testing and data storage.

Typically, LDVs have a dynamic range of 80-120dB- this again is high due to the lack of a mass/spring system, only limited by optical signal strength. They operate at between around 0-40°C, limited because they use equipment such as lasers and photodetectors that are temperature-sensitive. This level of technology also makes them very expensive- basic level LDVs can cost between £8,000-20,000, with top-range models costing upwards of £80,000.

Because of the lack of contact, LDVs are very lightweight. This makes them useful in vibration and structural dynamics measurement [6], mainly in situations such as:

- Testing mechanical structures
- Turbine blade vibration
- Aerospace structure testing
- Automotive vibration
- Data storage and electronics

Most of these require the structures and properties to remain unaffected, so the non-contact method of LDVs is perfect.

Strain gauge accelerometers

Strain gauge accelerometers measure acceleration by detecting strain in a beam attached to a proof mass. When it experiences acceleration, the inertial force ($F = ma$) from the proof mass causes the beam to bend, causing mechanical strain ϵ on the beam. Strain gauges bonded to the beam detect this strain as a change in electrical resistance. The change in relative resistance is:

$$\frac{\Delta R}{R} = G\epsilon$$

Where G is the gauge factor- this is typically ≈ 2 for metallic gauges and higher for semiconductor gauges [7]. The gauges are usually arranged in a Wheatstone bridge configuration [8]- this is an electrical circuit used to measure very small changes in resistance with high accuracy. It allows small resistance variations to be amplified into measurable voltages, making the strain gauge great for measuring low frequency vibration. It also cancels common effects like temperature change, improves sensitivity and noise rejection.

This arrangement converts strain into voltage as strain is proportional to applied force, which is also proportional to acceleration. This means the output voltage is linearly related to acceleration, something which is incredibly useful when taking readings. It means mechanical input is directly converted into an electrical signal without any distortion or changing any scales, something which is essential for vibration analysis and acceleration measurement.

Because their output doesn't rely on charge retention, strain gauges are capable of measuring static and low-frequency accelerations. They are commonly used where slowly varying or constant accelerations must be measured reliably, as unlike technologies like piezoelectric accelerometers, they can measure constant acceleration such as gravity. Their robustness makes them useful for shock measurement, however resistance changes due to temperature can introduce drift. They also have a limited response to higher frequencies due to their relatively high mechanical compliance [9]. Compliance is the measure of how easily a structure deforms when a force is applied, and is the inverse of stiffness:

$$\text{Compliance} = \frac{1}{\text{Stiffness}}$$

A high compliance structure bends or deflects easily- just like a strain gauge accelerometer. They rely on a small beam to bend, meaning that because the sensing element must bend to create measurable strain they therefore have a higher compliance. Because higher

compliance decreases stiffness, it therefore decreases natural frequency (f_n) as seen in equation (3), where k = stiffness and m = mass.

A lower natural frequency means a lower usable measurement bandwidth, meaning high frequency capability is limited. This means strain gauge accelerometers can measure frequencies from DC (0Hz) up to around 5kHz, with some custom sensors achieving up to 15-20kHz.

They rely on resistance change, giving the gauge a wide dynamic range (80-110dB) and straightforward amplification/filtering. They operate over a wide range of temperatures, from -50°C - 120°C , and are quite lightweight, weighing from $<10\text{g}$ - 100g depending on the enclosure and mounting.

Performance also affects cost, with strain gauge accelerometers costing anywhere from £100 for entry level models to £5,000+ for top of the range ones.

COMPARISON OF VARIOUS TECHNOLOGIES

The 4 vibration measurement technologies reviewed have different advantages and limitations, each with different applications.

Frequency bandwidth is an important value when it comes to picking which to use. Piezoelectric accelerometers and LDVs have an extremely wide bandwidth, making them ideal for high-frequency machinery diagnostics and modal analysis. On the other hand, MEMS and strain gauge accelerometers are limited to low-to-moderate frequencies, meaning if high-frequency components or capturing fine structural resonances is required, piezoelectric or LDV systems are preferred over MEMS/strain gauge.

With sensitivity and dynamic range, piezoelectric accelerometers offer high sensitivity and dynamic range, making them suitable for detecting small amplitude vibrations in noisy environments. LDVs, with their non-contact optical measurement, can achieve very high dynamic ranges for most purposes and are highly sensitive even at very low amplitudes. MEMS accelerometers have lower sensitivity and a smaller dynamic range (80-120dB), making them less appropriate for very subtle or high-frequency vibration measurements. Strain gauge accelerometers have moderate sensitivity and dynamic range, and are ideal for capturing static or low-frequency motion, e.g. tilt or slow structural shifts, which piezoelectric accelerometers can't measure due to their inability to measure DC.

Cost and mass also influence technology selection. MEMS accelerometers are extremely small, lightweight and cheap making them perfect for large scale manufacture of mobile phones and wearable electronics. Piezoelectric and strain gauge accelerometers are heavier and more expensive, with masses typically tens of grams and costing from hundreds to thousands of pounds due to the quality and quantity of components used. LDVs are the heaviest and most expensive, costing over £10,000 for research-grade instruments, which restricts them to being used only in laboratories or critical testing applications.

In terms of measurement, strain gauge accelerometers

are uniquely capable of measuring static and low-friction acceleration, making them ideal for structural monitoring, tilt measurement or crash testing. MEMS accelerometers also measure DC but are limited in dynamic range and resolution compared to strain gauges. Piezoelectric accelerometers have stiff sensing elements, which combined with their inability to detect DC makes them favoured in high-frequency or dynamic environments where their high linearity and low noise are useful. LDVs use non-contact measurement, allowing the measurement of fragile, rotating or hot surfaces without the effects of adding mass.

In summary:

- Piezoelectric- High frequency response and dynamic range, can't measure DC, moderate cost, suitable for machinery & structural dynamics
- MEMS- Low cost, small size, measures DC, lower sensitivity & bandwidth, ideal for consumer electronics and automotive sensors.
- LDV- Non-contact, high bandwidth & sensitivity, expensive and sensitive surface reflectivity, best used in lab testing and rotating machinery
- Strain gauge- Measures DC, robust, limited bandwidth & dynamic range, best used for tilt measurement, static acceleration or slow structural vibrations.

RECOMMENDATION FOR AUTOMOTIVE ENGINE APPLICATION

"I would like to measure the vibration of an automotive engine"

Automotive engines usually produce vibrations with frequencies ranging from 10Hz-5kHz, with the higher frequencies (10-15kHz) being caused by gear interactions or bearing defects. Piezoelectric accelerometers are able to measure from 1/2Hz up to 10-20kHz, covering the full range required. MEMS only cover up to 2-5kHz, restricting their ability to measure the higher end of the frequency bandwidth of the automotive engine. Strain gauge accelerometers, despite being able to measure DC, have a lower bandwidth and higher mechanical compliance, also reducing their ability to measure higher-frequency vibrations. LDVs offer an extremely wide bandwidth, however their cost and size makes them financially inefficient.

The engine vibrations vary from a few milli-g when idle (around -50dB), to tens/hundreds of milli-g (around 40dB) when in transient conditions. Piezoelectric accelerometers are available with ranges from $\pm 10g$ to $\pm 500g$, while still having dynamic ranges of 120-140dB, allowing the capture of both low-level vibration and high peaks when in transient conditions. MEMS accelerometers usually have dynamic ranges of 80-100dB, which might limit the resolution of readings of small vibrations when the engine is undergoing large accelerations. Strain gauge accelerometers have a similar dynamic range (80-110dB) but at the expense of bandwidth. Therefore, piezoelectric accelerometers are

again the best signal and noise wise across the amplitude spectrum

In terms of environmental sustainability, automotive engines have high temperatures, oil contamination and high mechanical shock. Industrial piezoelectric accelerometers can be bought which have sealed housings and are constructed with high stiffness. MEMS accelerometers, although they're compact, are less robust at constant high temperatures unless specially packaged for use in the engines. LDVs avoid contact with the hot surfaces, avoiding contact issues, but they need optical access and stable alignment, both of which are difficult to be maintained inside the engine bay.

Overall piezoelectric accelerometers are the best practical option. Their low mass, perfect fit for the requirements of the engine and relatively low cost makes them the ideal vibration measurement technology for measuring the vibration of an automotive engine.

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